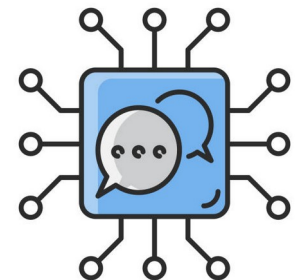


GRU and LSTM

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The "Goldfish" Memory

- How RNNs work: They process data sequentially, passing information from one step to the next.
- The Issue: Standard RNNs have a Short-Term Memory.
- The Analogy: Imagine reading a long book. By the time an RNN reaches the last chapter, it has completely forgotten who the main character is from Chapter 1.
- Technical Term: The Vanishing Gradient Problem. As information travels back through time during training, the signal gets weaker and weaker until it disappears.

Adding a "Brain" to the Network

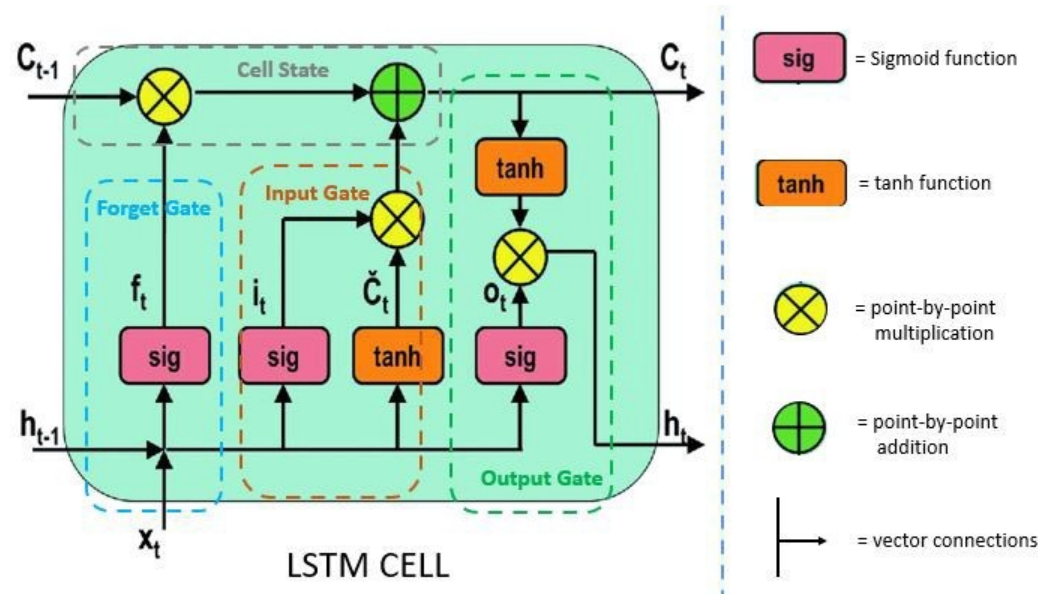
- The Fix: We need a neural network that chooses what to remember and what to forget.
- The Method: We introduce Gates.
- Analogy: Think of gates like a water faucet.
 - Open: Let information flow through.
 - Closed: Block information.
- The Heroes:
 - LSTM (Long Short-Term Memory)
 - GRU (Gated Recurrent Unit)

Long Short-Term Memory (LSTM)

- Invented: 1997 (Hochreiter & Schmidhuber).
- The Big Idea: LSTM adds a separate "super-highway" for information called the Cell State.
- The Cell State:
 - Information runs straight down the entire chain with minor interactions.
 - It's like a conveyor belt: easy for information to jump on and stay there unchanged for a long time.

LSTM Networks

- Every LSTM network basically contains three gates to control the flow of information and cells to hold information. The Cell States carries the information from initial to later time steps without getting vanished.



The Three Gates of LSTM

- LSTM uses 3 distinct gates to control the Cell State:
 - Forget Gate: Decides what information to throw away from the past.
 - Input Gate: Decides what new information to store in the cell state.
 - Output Gate: Decides what part of the cell state to output to the next step.

Step 1: The Trash Can (Forget Gate)

- Goal: Clean up the memory.
- Scenario: You are reading a text. You see "He eats." The network remembers the subject is "Male."
- The Change: The text changes to "She runs."
- The Action: The Forget Gate tells the cell state: "Forget the 'Male' gender information, it is no longer relevant."
- Math: It uses a Sigmoid function (outputs 0 or 1). 0 = Forget completely, 1 = Keep completely.

Step 2: The Notebook (Input Gate)

- Goal: Save new important information.
- Scenario: Continuing from "She runs."
- The Action: The Input Gate analyzes the new word ("She") and decides to store the new gender info ("Female") into the cell state.
- Process:
 - Decide which values to update.
 - Create a vector of new candidate values.
 - Add this to the Cell State.

Step 3: The Speaker (Output Gate)

- Goal: Provide the relevant info for the immediate prediction.
- Scenario: The network needs to predict the next word.
- The Action: Based on the current cell state ("Female" + "Running"), the Output Gate decides the next word should probably be a verb form or an adjective related to her.
- It filters the cell state and sends the "Hidden State" to the next time step.

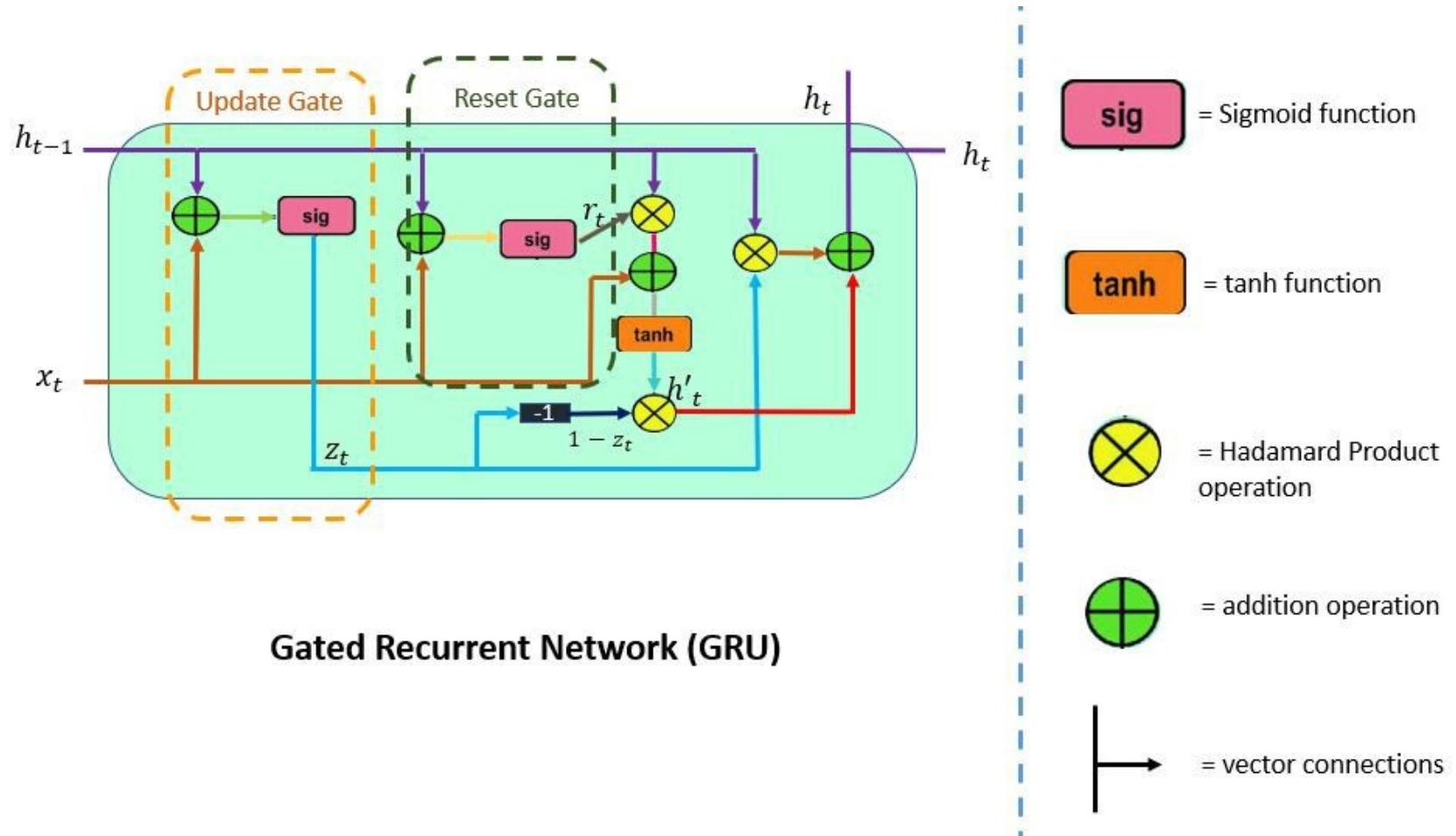
Gated Recurrent Unit (GRU)

- Invented: 2014 (Cho et al.).
- The Concept: A streamlined, more efficient version of the LSTM.
- Analogy: If LSTM is a fully loaded luxury car with many buttons, GRU is a sports car—lighter, faster, but still powerful.
- Key Difference: It does not have a separate Cell State. It merges the Cell State and Hidden State into one.

The Two Gates of GRU

- GRU simplifies the architecture to just 2 gates:
 - Reset Gate: Decides how much of the past information to forget. (Short-term focus).
 - Update Gate: A combination of the LSTM's Input and Forget gates. It decides how much of the past to keep and how much new info to add.

GRU



Gated Recurrent Network (GRU)

The Two Gates of GRU

Feature	LSTM	GRU
Complexity	High (3 Gates)	Low (2 Gates)
Speed	Slower to train	Faster to train
Data Size	Better for large datasets	Better for smaller datasets
Accuracy	Often slightly higher power	Often matches LSTM efficiency
Analogy	Heavy-duty Truck	Sports Car

Summary

- Standard RNNs fail at long sequences (Vanishing Gradient).
- LSTMs fix this using a Cell State and 3 Gates (Forget, Input, Output) to preserve long-term dependencies.
- GRUs are a simplified version with 2 Gates (Update, Reset) that merge the cell and hidden states for efficiency.
- Both are the backbone of modern NLP (before Transformers arrived) and Time-Series analysis.

Examples:

- Practical

Thank you

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